

DU TUAN AN

Manufacturing & Industrialization Engineer | Product Engineering | NPI | Tooling | AI-Assisted Engineering

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Purpose: provide representative, recruiter-verifiable evidence behind the CV without publishing a complete archive of employer engineering data. Internal drawings are cropped/sanitized; commercially sensitive details are omitted. Original files are retained privately where available.

~150 SKU variants - Huaqing CV scope	20+ OEM suppliers evaluated	22 / 14d Jintai SKUs delivered
15 / 7d Huaqing SKUs rebuilt	2 / 2 IKEA GO/NOGO passed	Up to 5x Selected automated tasks

What this portfolio verifies

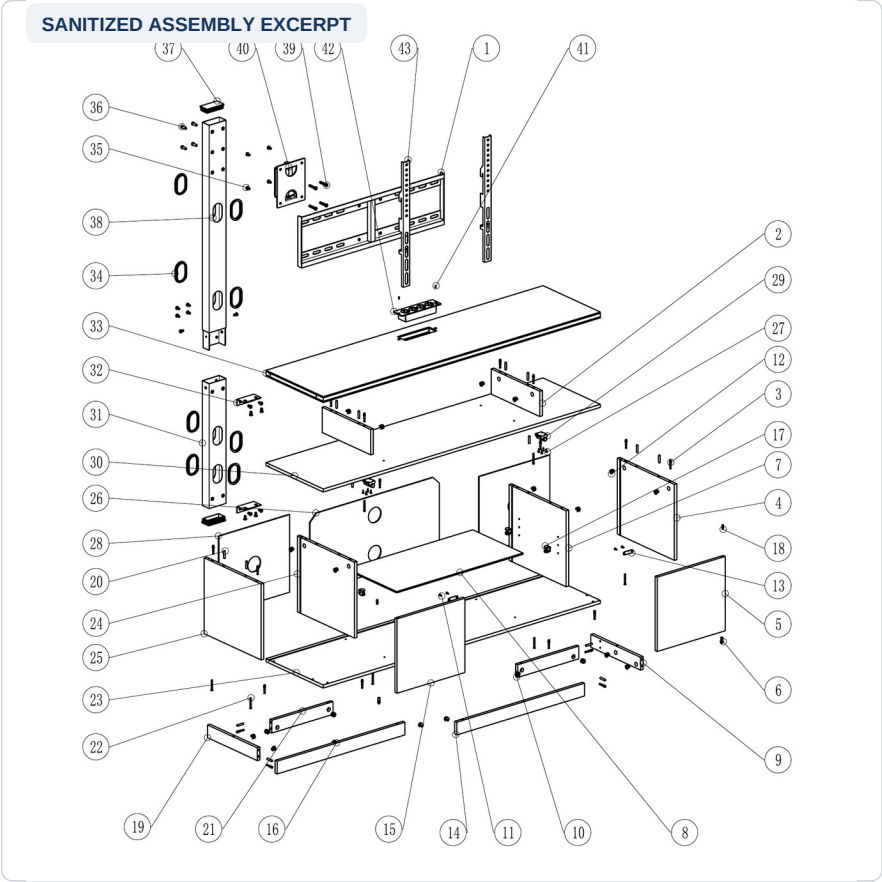
- **Direct engineering authorship:** retained title blocks naming designer / revision owner (candidate name shown in the Chinese title block).
- **Structural engineering depth:** production drawings with Q195 steel, welding, threaded features, tolerance and coating requirements.
- **Manufacturing implementation:** Siemens NX equipment/jig concepts paired with fabricated shop-floor hardware.
- **Current product engineering:** SONGMICS drawings at Jintai naming the candidate as designer and revision owner.
- **Digital engineering:** public GitHub repositories for AI-integrated PDM, NX automation and Google x Kaggle agent capstone.

Verification legend

DIRECT	Title block / authored source	PHYSICAL	CAD to fabricated equipment	SOURCE	Retained editable/internal	PUBLIC	Public GitHub source
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Product-Level Ownership & Lifecycle

Representative assembly evidence from a large OEM/ODM portfolio



DIRECT AUTHORSHIP									
HUAQING									
材料		规格	单位	数量	设计	审核	批准	日期	
颜色		颜色	比例	1:12	设计	审核	批准		
设计重量		kg	投影		审核	批准			
品牌		公差标准	公差标准	SS-130-2708-1	批准	批准			

Title-block verification. The retained assembly package shows HUAQING documentation with the candidate listed in the design/revision chain. Product codes and sensitive identifiers are masked in this public version.

Scope represented
Assembly structure, BOM-linked parts, dimensions, revision control and supplier-facing engineering release.

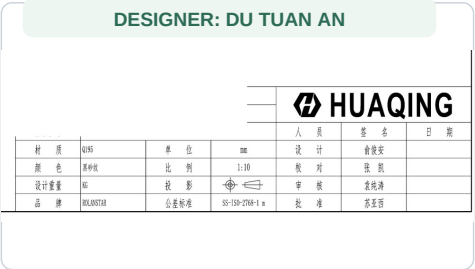
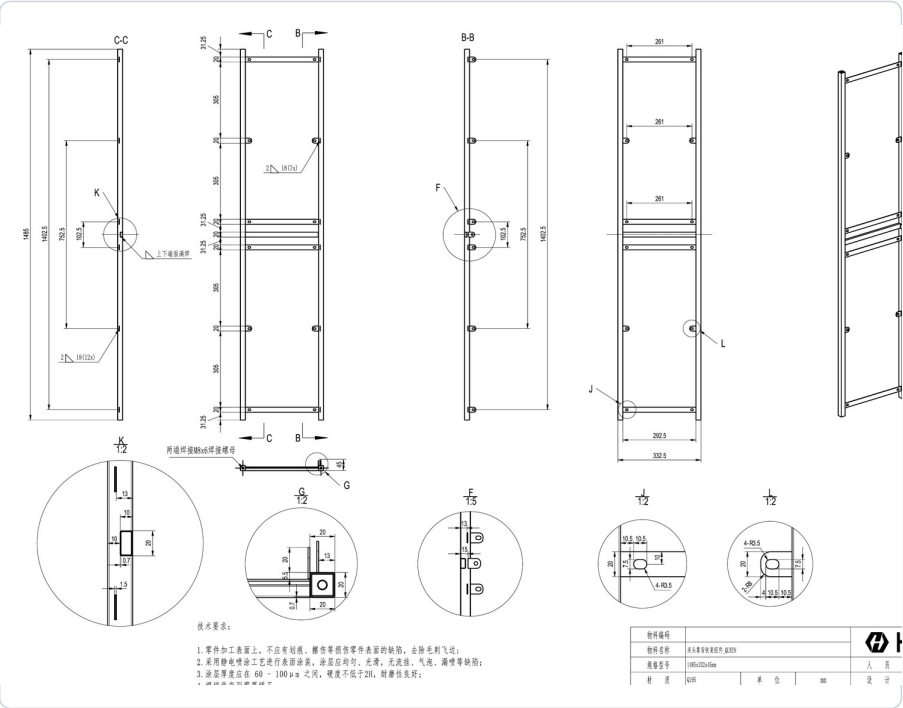
Evidence supported by retained files

- Exploded assembly drawings and overall dimensional drawings across multiple product/SKU variants.
- Revision-controlled title blocks with separate designer, checker, reviewer and approver roles.
- A retained planning index covering the product set handled during the role; the CV uses the conservative claim of nearly 150 SKU variants.
- Representative product packages connect assembly drawings, detailed components, BOMs and downstream documentation.

Interpretation for recruiters: this page is not intended to expose the full employer archive. It establishes that the candidate worked at product/assembly level with revision-controlled engineering documentation rather than only producing isolated CAD geometry.

Detail Structural Engineering

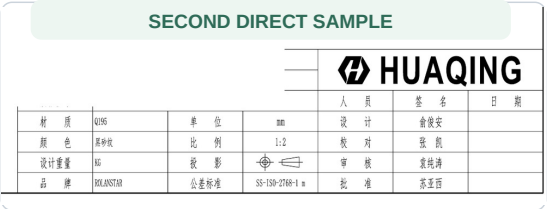
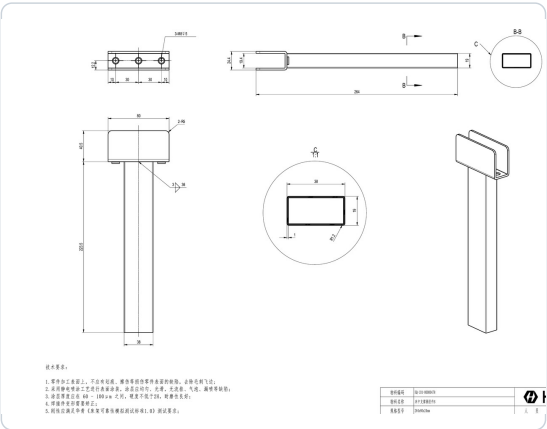
Representative production drawings - welded steel furniture components



What the drawings demonstrate

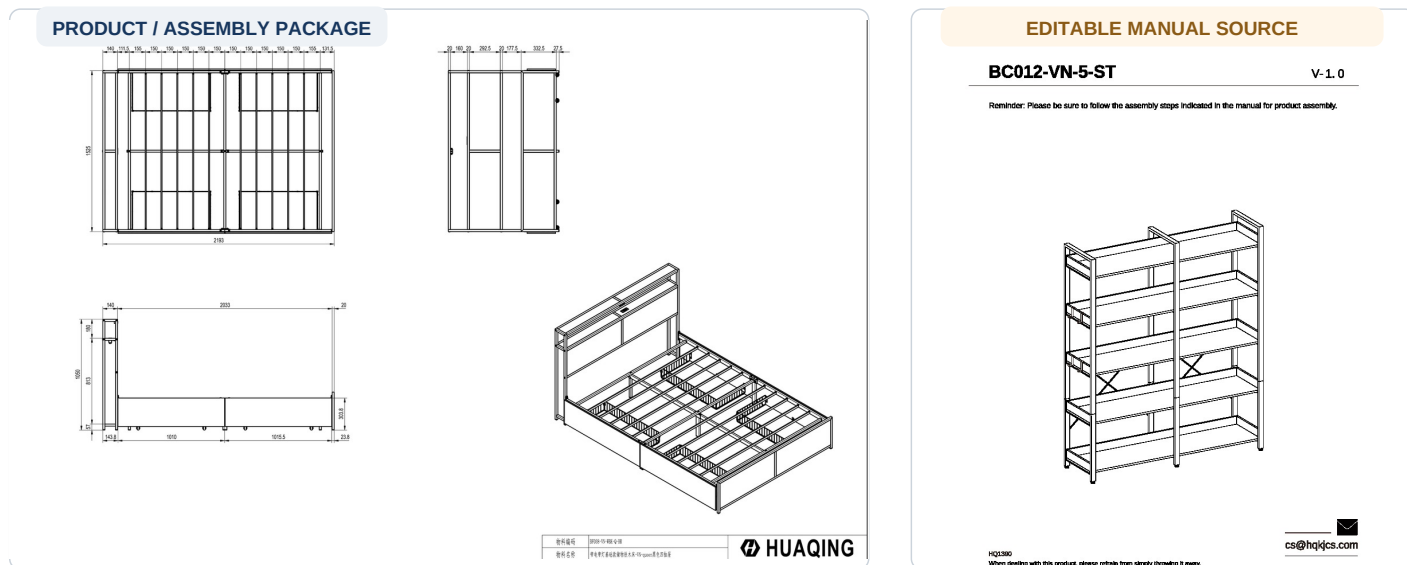
- **Material:** Q195 steel with black powder-coat finish.
- **Manufacturing definition:** welded assemblies, welded nuts, threaded M8 features, radii, sections and localized detail views.
- **Process requirements:** weld-deformation correction, burr removal and electrostatic coating controls.
- **Quality / tolerance:** SS-ISO-2768-1 m general tolerance; coating thickness 60-100 μm and hardness requirement $\geq 2H$ on retained component drawings.
- **Functional requirement:** rigidity to meet Huaqing bed-frame reliability simulation/test standard.

A second retained component drawing independently repeats the same designer name and production requirements, strengthening authorship evidence beyond a single title block.

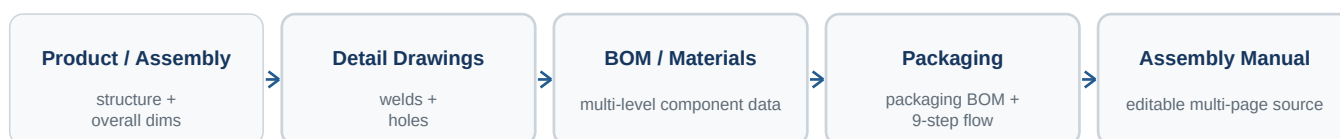


Engineering Lifecycle & Documentation

From product structure to manufacturing and assembly documentation



Representative engineering-data chain



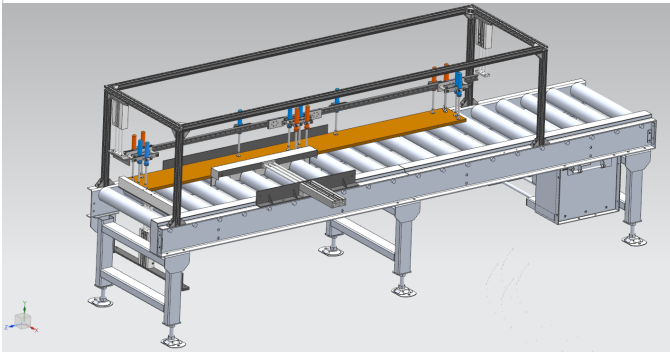
Retained documentation evidence

- A 3-page product package contains an exploded/overall product definition and a 44-line product BOM spanning welded steel assemblies, panels, hardware, lighting/power components, drawers and plastics.
- A separate retained workbook contains a full material/hardware hierarchy, packaging BOM and a 9-step packing sequence.
- An editable Inkscape/SVG assembly-instruction source contains 16 pages and export configuration for a production PDF manual. The source does not embed an author field; it is presented as supporting evidence within the wider product-ownership chain.
- Raw BOM codes, supplier/commercial fields and other reconstructable details are intentionally not reproduced in this public portfolio.

Manufacturing Automation & Tooling

Siemens NX concepts paired with fabricated shop-floor equipment

3D CAD - NX

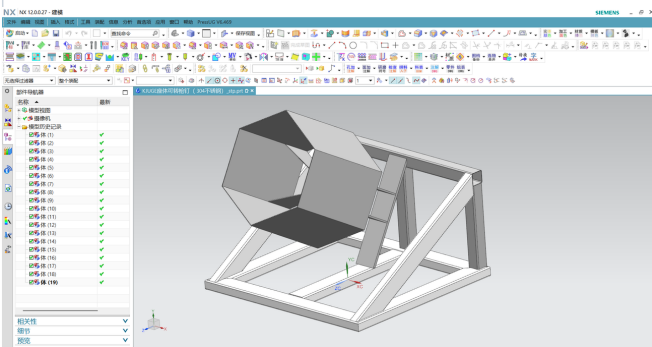


FABRICATED EQUIPMENT



Hole-processing / punching equipment. Large-format roller/workstation concept with distributed pneumatic/fixture heads. The public photo is background-blurred to avoid exposing coworkers.

3D FIXTURE CONCEPT



SHOP-FLOOR IMPLEMENTATION

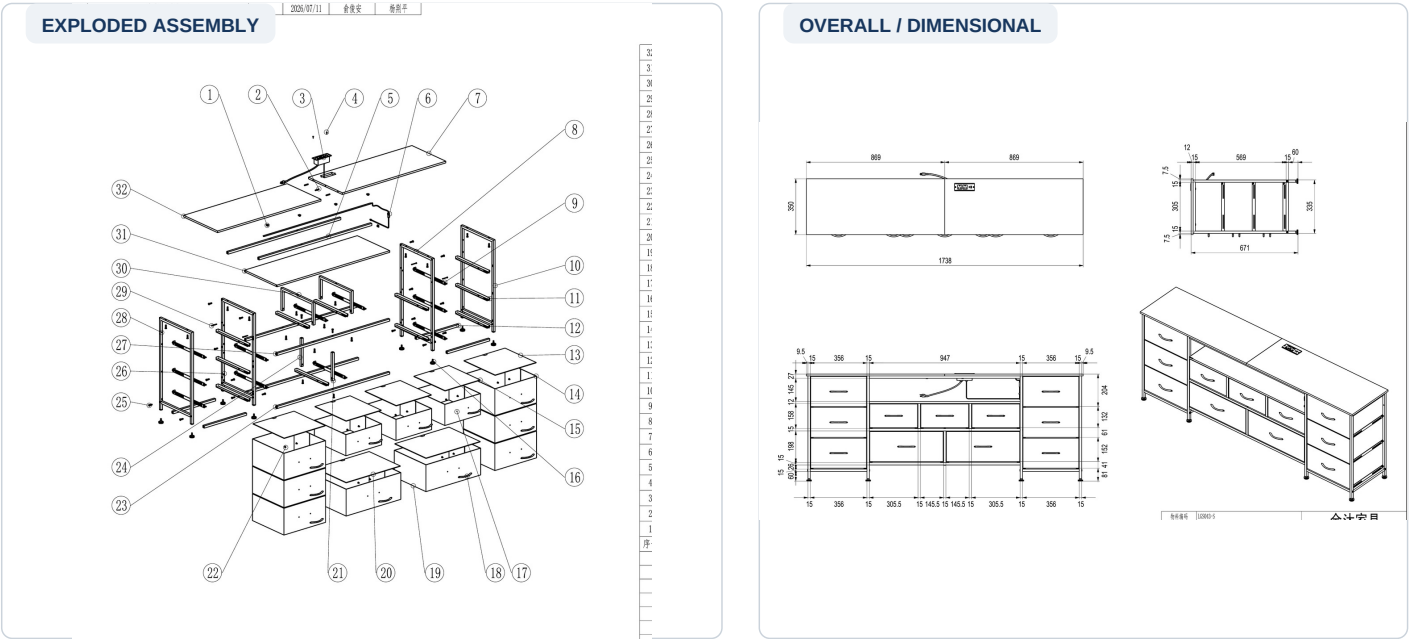


Positioning / rotation fixture. A separate NX concept is paired with fabricated hardware used on the production floor. Additional retained photos show multiple fabricated jigs/fixtures from the same period.

Evidence boundary: retained material verifies 3D design concepts and fabricated equipment. Original 2D machine-detail drawings are not retained, so this portfolio does not claim to present a complete machine manufacturing package. Supported role: mechanical concept, 3D equipment/fixture design, fabrication follow-up and shop-floor implementation.

Current Product Development

SONGMICS product engineering - representative revision-controlled drawings



Direct evidence

- Jintai title block names the candidate as designer on a SONGMICS product drawing (candidate name shown in the Chinese title block).
- Revision record identifies the candidate as the change owner for V3.1 on the retained product package.
- The drawing package combines exploded assembly, structured parts list and overall product dimensions.

DESIGNER / REVISION OWNER						
				金汰家具		
				人 员	签 名	日 期
材 质	Q195APB&布抽	单 位	mm	设 计	金俊安	
颜 色	黑色/仿古色	比 例	1:10	校 对	赵刚	
设计重量	KG	投 影		审 核	赵刚	
品 牌	SONGMICS	公差标准	SS-ISO-2768-1 m	批 准	杨凯平	

Current-role delivery metrics (CV)

22 SKUs delivered in 14 days; CNC production files for product lines within 24 hours; new-product engineering / prototype optimization / customer sample approval within 3 days. Representative drawings are shown here; the exact batch-delivery archive is not reproduced publicly.

Portfolio links

GitHub profile: github.com/dutuanan96
LinkedIn: [linkedin.com/in/an-du-010b83420](https://www.linkedin.com/in/an-du-010b83420)

AI-Assisted Engineering Systems

Public source repositories - recruiter-verifiable implementation evidence

FLAGSHIP PROJECT

AI-Integrated PDM & Engineering Change System

github.com/dutuanan96/bom-viewer-sync

Browser-based Viewer/Admin PDM covering product, material, BOM, revision and engineering-document workflows. AI is evidence-first: canonical PDM retrieval, structured change proposals, before/after diff, domain/graph validation, risk warnings, engineer review and explicit final save.

PUBLIC SOURCE

Siemens NX12 Engineering Export Automation Dashboard

github.com/dutuanan96/nx12-export-dashboard

AI-assisted NX automation for batch PRT -> PDF, PRT -> STEP (AP214), PRT -> DWG and STP/STEP -> IGES. The retained source includes NX Journal workflows, output validation, log cleanup/retry, timeout handling and practical production safeguards.

PUBLIC SOURCE

Furniture Manufacturing AI Agent - Google x Kaggle Capstone

github.com/dutuanan96/furniture-agent-capstone

Google ADK multi-agent BOM assistant with a root coordinator, specialized BOM/comparison agents, tool-based memory and Excel export. The project uses deterministic BOM parsing/comparison and bilingual Vietnamese/Chinese workflows with read-only safeguards.

PUBLIC SOURCE

Role across digital projects: engineering requirements, domain/workflow design, AI-agent orchestration, testing, validation and deployment. Implementation was produced with AI coding agents; this portfolio does not claim traditional software-development expertise.

Evidence Map

How a recruiter can independently check the principal CV claims

CV / portfolio claim	Verification route	Evidence type
Huaqing product scope	Retained planning index + representative engineering packages	CORROBORATED
Drawing authorship	HUAQING title blocks naming designer / revision owner - candidate name shown in title block	DIRECT
Detail structural engineering	Q195 production drawings with welding, M8 features, tolerance & coating notes	DIRECT
BOM / packaging	44-line product BOM + full material/hardware and packaging workbook	PRIVATE SOURCE
Assembly instructions	Editable 16-page Inkscape/SVG manual source	SOURCE
Henglin tooling / automation	Siemens NX CAD paired with fabricated equipment photos	PHYSICAL
Jintai product engineering	SONGMICS drawings naming designer / change owner - candidate name shown in title block	DIRECT
AI PDM / NX automation	Public GitHub repositories linked in this portfolio	PUBLIC

Confidentiality note. This portfolio intentionally shows representative excerpts rather than a complete archive. Supplier names, commercial planning data, material codes, quantities and other reconstructable employer information are omitted or masked. Original records can be discussed privately during a legitimate interview where appropriate.

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